Great Salt Lake Shrinkage and Potential ML Applications

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Utah's Great Salt Lake is shrinking at a rapid rate. If Utah were to deplete its water resources, this would spell disaster for economic and ecological systems across the western United States. However, stemming water consumption is politically unpopular since many agricultural and industrial businesses rely heavily on water supplied by the lake; reducing consumption could adversely affect the job market and disrupt the agricultural sector that accounts for a large portion of Utah's GDP. To "fully restabilize the Great Salt Lake," statewide water consumption would have to be reduced "by more than half," which is a tall order considering agriculture alone consumes "74% of all the water that is being currently pulled from the system" (RealLifeLore, 2025). Using machine learning (ML) to simulate policy and planning scenarios for the Great Salt Lake crisis offers methods for foresight and decision-making, but it also introduces complex risks and trade-offs.

There are several ways ML can be leveraged to simulate how different policies and scenarios can affect water supply. ML can simulate how different water policies (e.g., agricultural restrictions, urban conservation, wetland restoration) affect lake levels, salinity, and ecosystem health over time. Lee et al. (2025) demonstrated that a hybrid cluster-based wavelet-ML approach with models trained on historical climate, hydrology, and usage data can forecast droughts, inflows, and ecological tipping points, giving policymakers lead time to act. By integrating economic, demographic, and land use data, ML can predict which counties would bear the brunt of water cuts or benefit from conservation subsidies. ML can also power dashboards that let policymakers and the

public explore “what-if” scenarios, which would open the door to more participatory policymaking.

However, these ML applications also bear some risk. Policymakers might treat ML outputs as gospel, ignoring uncertainties, edge cases, or ethical concerns. If policymakers do not consider potential flaws in the ML models, such as if historical data is inaccurate or reflects inequity, they risk making decisions that reinforce existing problems. Furthermore, sophisticated ML tools may centralize decision-making among technical elites, or worse, lead to general distrust of data scientists and inspire decisions completely divorced from scientific process. This risk is likely if the generated reports are unfavorable to the general public, such as if simulations suggest that water reallocations could devastate certain sectors such as alfalfa farming and trigger job losses. To navigate these challenges and use ML responsibly, we must ensure transparency about our models and AI techniques, co-design models with those who would be affected by water policy changes, explicitly encode equity and sustainability into model objectives, and deliberately use ML as a tool for insight instead of as an arbiter of truth.

References

Lee, S., Mehr, A. D., Moriasi, D., & Mirchi, A. (2025). Large-Scale drought forecasting in the U.S. southern plains through a hybrid Cluster-Based Wavelet-Machine learning approach. *AGU Journals*. https://doi.org/10.1029/2024WR039744

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